

後端範例 Backend Examples

Purpose / 目的：提供非執行性（non-executable）的資料庫與後端 API 結構範例，供 Designer GPT 在生成規格書時參考。

1. 實體與關係 Entities & Relationships

範例 Example:

```
User { id: UUID, email: string(unique), name: string, role: enum(admin|member), createdAt, updatedAt }
Event { id: UUID, name: string, startsAt: datetime, endsAt: datetime, capacity: int>=0, createdBy: User.id }
```

關聯 Relationships: - User (1) — (n) Event - Constraint: startsAt < endsAt; capacity >= 0

2. REST API 合約 REST API Contracts

```
GET /api/events?cursor&limit → 200 { items: [Event], nextCursor }
POST /api/events (auth: admin) → 201 { Event }
PATCH /api/events/:id (admin|owner) → 200 { Event }
DEL /api/events/:id (admin|owner) → 204
```

3. 驗證與錯誤格式 Validation & Error Shape

```
Validation: Zod/Joi-equivalent at API boundary
422 Validation Error: { errors: [ { path, msg, code } ] }
Generic Error: { error: { code, message, details? } }
```

4. 認證與安全 Auth & Security

- Auth: JWT (HS256) placeholder
 - Middlewares: requireAuth, requireRole('admin')
 - CORS: allow app origin only
 - HTTPS required in production
 - Rate limit: 100 req/min per IP on POST endpoints
 - Idempotency: header `Idempotency-Key` for POST creates
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5. 非功能需求 Non-functional

- Pagination: cursor (default limit=20, max=100)
- Observability: /healthz endpoint, request-id in logs
- Performance: indexed fields on Event(startsAt), User(email)

6. OpenAPI 範例片段 OpenAPI Snippet

```
openapi: 3.0.3
info: { title: Example API, version: 1.0.0 }
paths:
  /api/events:
    get:
      parameters:
        - in: query
          name: cursor
          schema: { type: string }
        - in: query
          name: limit
          schema: { type: integer, default: 20, maximum: 100 }
      responses:
        '200': { description: OK }
    post:
      responses:
        '201': { description: Created }
```

7. GraphQL 替代範例 GraphQL Alternative

```
type Event { id: ID!, name: String!, startsAt: String!, endsAt: String!, capacity: Int!, createdBy: ID! }
type Query { events(cursor: String, limit: Int! = 20): [Event!] }
type Mutation { createEvent(input: EventCreate!): Event! }
```

8. 測試用樣本資料 Sample Fixtures

```
{
  "users": [ {"id": "u_1", "email": "admin@example.com", "name": "Admin", "role": "admin"} ],
  "events": [ {"id": "ev_1", "name": "Launch
Day", "startsAt": "2025-10-06T09:00:00Z", "endsAt": "2025-10-06T11:00:00Z", "capacity":
120, "createdBy": "u_1"} ]
}
```

9. 使用說明 Usage Notes

- 本文件僅作為結構與命名參考，實際規格由應用情境導出。
- 生成代理必須嚴格遵守最終規格書中的欄位與格式。
- 預設技術棧：Node.js + Fastify/Express + PostgreSQL + Prisma。